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Course: Fundamentals in AI And ML

📩 Spam Message Detection System Using Machine Learning

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1. Introduction

In today's digital world, communication through messages has become an essential part of daily life. However, along with useful communication, there has been a rapid increase in unwanted and harmful messages known as spam. These messages often contain advertisements, fraudulent offers, or misleading information. Spam messages not only waste time but can also lead to serious issues like scams and data theft.

To address this problem, this project focuses on building a Spam Message Detection System using Machine Learning techniques. The system is designed to automatically classify messages as either spam or not spam (ham), helping users filter unwanted content efficiently.

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2. Problem Statement

The increasing volume of spam messages makes it difficult for users to manually identify and filter them. Traditional filtering methods are not always effective as spam messages keep evolving in style and content. Therefore, there is a need for an intelligent system that can automatically detect spam messages based on their textual content.

3. Objectives

The main objectives of this project are:

- To develop a system that can classify messages as spam or not spam
- To implement machine learning techniques for text classification
- To provide a simple and user-friendly interface for users
- To understand the basics of Natural Language Processing (NLP)

4. Methodology

The project follows a structured approach to build the spam detection system:

4.1 Data Collection

A small dataset was manually created containing example messages labeled as spam or ham. This avoids dependency on external sources and ensures simplicity.

4.2 Data Preprocessing

The text data is cleaned and prepared for analysis. This includes: - Removing unnecessary symbols - Converting text into a usable format

4.3 Feature Extraction

The textual data is converted into numerical form using Count Vectorization. This process transforms words into vectors so that machine learning algorithms can process them.

4.4 Model Training

A Naive Bayes algorithm is used to train the model. This algorithm is widely used for text classification problems because of its simplicity and effectiveness.

4.5 Prediction

Once trained, the model predicts whether a given message is spam or not based on learned patterns.

5. Tools and Technologies Used

- Python
- Pandas (for data handling)
- Scikit-learn (for machine learning)

- Streamlit (for user interface)

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6. Implementation

The implementation of the project is divided into two parts:

6.1 Model Development

The dataset is loaded and split into input (text) and output (label). The text is converted into vectors using CountVectorizer. Then, a Naive Bayes classifier is trained on the data.

6.2 User Interface

A simple web interface is created using Streamlit. Users can enter a message, and the system will display whether it is spam or not.

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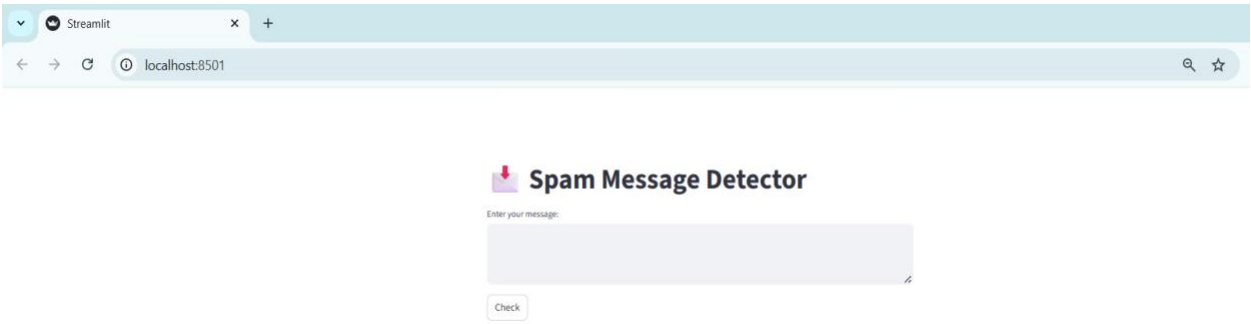
7. Results and Output

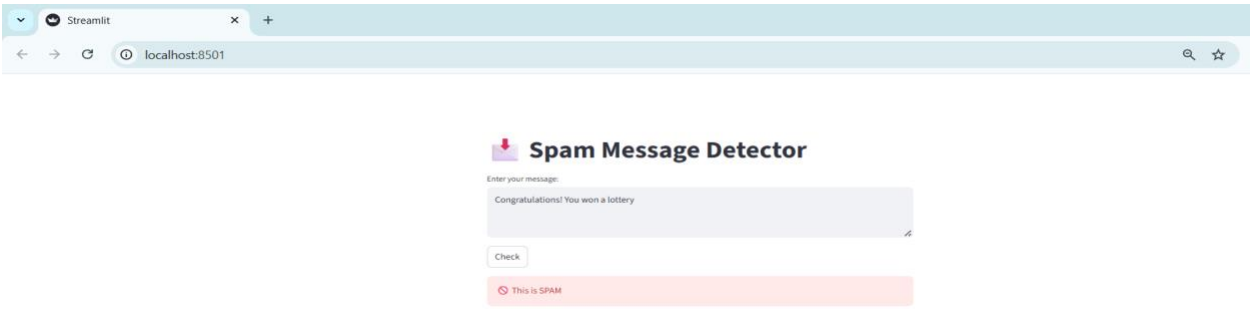
The system successfully classifies messages based on the trained model. When a spam message is entered, the system displays it as spam, and for normal messages, it shows not spam.

The results are displayed instantly, making the system user-friendly and efficient.

App interface:



Spam result:



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8. Challenges Faced

During the development of this project, a few challenges were encountered:

- Small dataset leading to limited accuracy
- Difficulty in handling unseen words
- Initial setup issues with Streamlit

These challenges were addressed through testing and debugging.

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9. Conclusion

The Spam Message Detection System successfully demonstrates how machine learning can be applied to solve real-world problems. The project highlights the effectiveness of simple algorithms like Naive Bayes in text classification tasks.

It also provides a basic understanding of Natural Language Processing and how text data can be transformed into useful insights.

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10. Future Scope

The project can be further improved by:

- Using a larger dataset for better accuracy
- Implementing advanced algorithms like Deep Learning
- Integrating real-time message filtering systems
- Enhancing the user interface with additional features

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11. References

- Scikit-learn Documentation
- Python Official Documentation
- Streamlit Documentation

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12. Acknowledgement

I would like to express my gratitude to my faculty and institution for providing the opportunity to work on this project. This project helped me gain practical knowledge in machine learning and improved my problem-solving skills.

✅ End of Report